



Territory Briefing: Massa-Carrara

Professional province-level screening based on ARCUS historical evidence, territorial exposure, ARCUS/AINOP Collapse Rate and historical-pattern reading.

PRIORITY

71/100

COLLAPSE

4.11x / national rank 16

EVENTS

7

DECISION MESSAGE

Concentrated flood scenario

REPORT DATE

Jun 10, 2026

CONTEXT

Bridge

PROVINCE

Massa-Carrara

OUTPUT

New construction

ARCUS is built on the first systematic database of collapsed bridges in Italy from 2000 to today: the report does not merely rank cases, it reads documented historical patterns to guide preliminary checks.

The selected province requires preliminary screening for Bridge interventions. The value is not the case list: it is understanding whether Massa-Carrara has already shown concentrated vulnerability under extreme conditions or risk distributed over time.

RECOMMENDED USE

ARCUS reading: for new Bridge interventions, the Hydraulic signal does not define a design solution. It identifies the technical attention fields that should be checked before design, due diligence or site-specific investigation decisions in the province of Massa-Carrara.

PRIORITY INDEX

71/100

exposure index + heritage reading

COLLAPSE RATE

4.11x

ARCUS/AINOP national benchmark; confidence high

HISTORICAL EVENTS

7

Hydraulic dominant

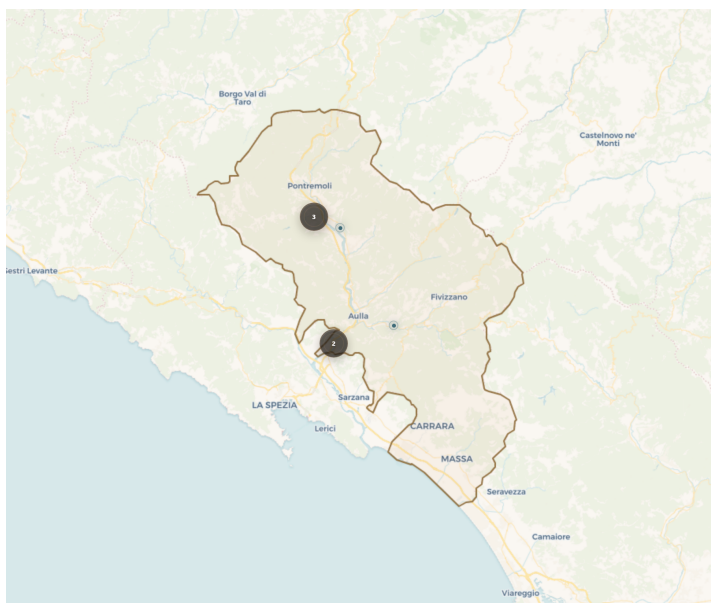
RELIABILITY

79/100

16 linked sources

COLLAPSE RATE INTERPRETATION

The 4.11x rate versus the national average places Massa-Carrara among the provinces with documented historical vulnerability above the ARCUS/AINOP benchmark. Massa-Carrara is national rank 16 by documented ARCUS/AINOP collapse rate. Collapse Rate confidence: high. The AINOP denominator should be read as available bridge-census coverage, not as a perfect inventory of the bridge stock.



MASSA-CARRARA / ARCUS ATLAS EXTRACT

GEOMORPHOLOGY AND CONTEXT

Massa-Carrara shows a hydraulic profile that should be read by geomorphological setting, not as a uniform signal: 7 ARCUS events across 5 municipalities indicate that confined or torrential watercourses, valley crossings and lowland river reaches must be separated during verification.

MAP READING

The map locates 7 ARCUS cases across 5 municipalities and should be read with the historical pattern: Concentrated flood scenario. Dominant date: 2011-10-25.

HISTORICAL PATTERN

Concentrated flood scenario. collapses concentrated in a single flood/extreme-hydraulic scenario (2011-10-25, 57%). The 4 cases concentrated on 2011-10-25 should be read as an event cluster, not as independent recurring events.

DOMINANT TRIGGER**86%**

Hydraulic

HAZARD**hydraulic**

dominant territorial signal

READING**single event**

57% in one flood scenario; check extreme single-event design scenarios.

- Use the territorial exposure indicators to identify the dominant Hydraulic signal.
- Review the documented concentrated historical scenario and use the map/appendix only as case reference.
- For hydraulic-triggered collapses in Massa-Carrara, priority data includes: hydraulic model with TR100/TR200 flood scenarios, preliminary scour depth estimate at piers, riverbed/bathymetry evidence at crossings and debris-transport indicators where mountain, confined-channel or minor-torrent basins are involved.
- Differentiate checks by geomorphology in Massa-Carrara: torrential, confined or steep watercourses require scour susceptibility and debris-transport checks; lowland river crossings require flood level verification, inundation duration and backwater sensitivity.
- Use CSV and GeoJSON exports for technical coordination.

DATA REQUEST PACKAGE

TR100/TR200 hydraulic model; preliminary scour depth; riverbed/bathymetry evidence; debris transport indicators where relevant.

COLLAPSE RATE CONFIDENCE NOTE

Collapse Rate confidence: high. The AINOP denominator should be read as available bridge-census coverage, not as a perfect inventory of the bridge stock.

The following list is a documentary reference, not an operational P1/P2/P3 ranking. The decision message remains the provincial historical pattern.

ID	Date	Municipality	Severity	Cause
B00.11.01	Nov 13, 2000	Filattiera	Total Collapse	Hydraulic
B11.10.02	Oct 25, 2011	Aulla	Total Collapse	Hydraulic
B20.04.02	Apr 08, 2020	Albiano Magra	Total Collapse	Landslide
B11.10.03	Oct 25, 2011	Aulla	Total Collapse	Hydraulic
B11.10.04	Oct 25, 2011	Castagnetoli	Total Collapse	Hydraulic
B12.11.01	Nov 11, 2012	Aulla	Total Collapse	Hydraulic
B11.10.08	Oct 25, 2011	Pontremoli	Partial Collapse	Hydraulic

Layer	Meaning
Priority Index	Territorial exposure index: synthesizes dominant hazard, ARCUS case density, event concentration and evidence reliability. It does not directly measure historical bridge-stock vulnerability.
Collapse Rate	Ratio between ARCUS cases and AINOP counted bridges in the province. Value: 4.11x, national rank 16; it should be read separately from the Priority Index.
Why Separate	Priority Index indicates where the territory has elevated hazard exposure; Collapse Rate indicates where the bridge stock has shown historical vulnerability above or below the national average.
Historical Failure Context	Historical-pattern reading: concentrated event or distributed risk over time.
Hydraulic Cause Category	Hydraulic cause category: ARCUS uses 'hydraulic' as the documented trigger when sources confirm a flood or high-water event as the proximate cause. Mechanism-level classification - scour, bank erosion, debris impact, overtopping - is applied only where primary technical sources provide
Evidence Window	Observed period: ARCUS database of collapsed bridges in Italy from 2000 to today; close events in the same time window are read as one scenario, not decision duplicates.
AINOP Coverage	Collapse Rate uses AINOP counted bridges as provincial denominator. Confidence: high; local undercounting may inflate the ratio.
Evidence Classes	A = primary institutional/technical source; B = reliable technical or media source; C = generic documentary evidence; D = weak evidence to strengthen.